



Universitat de Lleida

1st Project

Automated Reasoning and Learning

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1. `similar_genre("Fantasy", "Adventure").`
2. `similar_genre("Crime", "Drama").`
3. `similar_genre("Romance", "Comedy").`
4. `similar_genre("SciFi", "Fantasy").`
5. `series("edunai4", ["Comedy"], 14).`
6. `series("TheWire", ["Action", "Crime"], 5).`
7. `series("edunai5", ["Crime", "Action"], 7).`
8. `series("edunai7", ["Fantasy"], 2).`
9. `series("RedDwarf", ["Comedy", "SciFi"], 10).`
10. `series("edunai6", ["Drama", "SciFi"], 1).`
11. `series("edunai8", ["Fantasy"], 3).`
12. `series("MoonLighting", ["Romance", "Crime"], 5).`
13. `series("edunai3", ["Comedy"], 10).`
14. `series("edunai9", ["Adventure"], 3).`

```

matchGenre(S1,S2,CGenre):-
    series(S1, Genres1,_),
    series(S2, Genres2,_),
    S1 \= S2,
    member(CGenre, Genres1),
    member(CGenre, Genres2).

similarNumber(N1,N2,Dist):-
    X is abs(N1 - N2),
    X =< Dist.

matchGenreAndSeasons(S1, S2, CGenre, Dist):-
    matchGenre(S1,S2,CGenre),
    series(S1, _, N1),
    series(S2, _, N2),
    similarNumber(N1,N2,Dist).

matchSimGenre(S1, S2, G1, G2) :-
    series(S1, Genres1, _),
    series(S2, Genres2, _),
    member(G1, Genres1),
    member(G2, Genres2),
    similar_genre(G1, G2).

recSimGenre(S1,S2,G1,G2) :-
    matchSimGenre(S1,S2,G1,G2) ;
    series(S1,Genre1,_),
    series(S2,_,_),
    series(SI,_,_),
    member(G1, Genre1),
    similar_genre(G1,GI),
    recSimGenre(SI,S2,GI,G2).

```

Problem 1

Give the complete set of answers to the query:

```
matchGenreAndSeasons("TheWire", S2, CGenre, 4)
```

$(S2, CGenre) = \{(edunai5, Action), (edunai5, Crime), (Moonlighting, Crime)\}$

Problem 2

Give the complete set of answers to the query:

```
matchSimGenre("MoonLighting", S2, G1, G2)
```

$(S2, G1, G2) = \{(edunai4, Romance, Comedy), (RedDwarf, Romance, Comedy), (edunai6, Crime, Drama), (edunai3, Romance, Comedy)\}$

Problem 3

For the following query:

```
recSimGenre("RedDwarf", S2, "SciFi", G2)
```

Give the complete set of answers:

$(S2, G2) = \{(edunai7, Fantasy), (edunai8, Fantasy), (edunai9, Adventure), (edunai9, Adventure)\}$

Give the complete SLD resolution tree:

You can find the SLD resolution tree appended at the end of the document.

